Process Control: Level Control

Final Lab Report
Unit Operations Lab 6
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Group 2
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1. Executive Summary

The purpose of the experiment is to observe the effects of disturbances in a process control system for level management. This is accomplished via the Armfield PCT50 Level Control Process apparatus. There are two parts of the experiment in which a solenoid valve is either manually manipulated or automatically varied with the integrated software. The study is done to simulate scenarios of either small or big disturbances within a closed process system. The theory of the first section, experiment A, is the proportional relationship of the pump speed with the flowrate of valve CV2. The trend is confirmed in the experiment as the pump speed is constant at 35% pump power. Once valve CV2 is opened, the constant power is not able to maintain the level. The second part of experiment A is increased to 60% pump power in order to raise the level to 150 mm and took 56 minutes. When the power is increased to 75% power, it took 6 minutes to fill to 150 mm. The second section, experiment C, the control of the proportional band and the integral term is observed. This trend is also confirmed in the experiment as the higher the proportional band, the less responsive the system is to disturbances. When the value is set to P = 200%, there was no change until it was changed to 50% as the percent error is reduced. The integral term is manipulated to assist the value of P and offset response time. The trend of the theory is confirmed as stability of level is achieved with low level of integral term and proportional band relationship. Value of P that is successful is 20% at 100 seconds as there is faster responsiveness to disturbances. Finally, the process control is set to automatically change power with varying proportional variance from 100% to 2%. As expected, lower values of P results in higher oscillations of power as the software is trying to respond to small disturbances.

2. Introduction

Process control plays an integral role in chemical engineering because it assures that processes are running safely and efficiently while meeting all required specifications. Correctly controlling a process can optimize production and minimize safety hazards, which will economically benefit the company and avoid downtime for the plant. The controls implemented on a process continuously monitor operation conditions and make appropriate adjustments to ensure that the product and equipment specifications are not compromised.[1] This experiment focuses on level control. Level control is implemented anytime an interface exists between two phases, such as vapor and liquid, and is usually associated with equipment such as tanks, reactors, distillation columns, and drums. It is important to control level in various equipment to prevent flooding, drying out, or even pressure build-up.[2]

The objective of the experiment was to learn the fundamentals of process control by controlling the level of water in a process vessel and to determine the effect of P and I controllers on the system. In this experiment the apparatus used was the Armfield PCT50 Level Control process unit and software, which allowed for changes and disturbances to be quickly corrected and investigated.[3] The software used in the experiment continuously recorded the data which was later analyzed. Two different exercises were performed, where the first exercise relied on manually controlling the water level and the other exercises utilized automatic controls to maintain the level of the water.

3. Theory

Open Loop

The open loop control system, also known as a non-feedback system, is defined as being a system in which the output has no influence or effect on the control action of the input signal.[4] In other words, the control system is controlled manually and does not correct for errors or disturbances itself. Open loops take far longer to correct for disturbances and may not even alert when a disturbance occurs, which calls for the process to be supervised. In the experiment, the open loop system required for the level of the water to be controlled manually be directly changing the speed of the pump. The pump speed and the flow of the water were proportional meaning that if the pump speed was increased the level of the water would increase and if the pump speed was decreased the level of the water would also be decreased.

Closed Loop

The closed loop control system, also known as the feedback control system, is defined as being a system designed to automatically achieve and maintain desired output conditions.[4] An automated control loop has the ability to detect disturbances or errors through the use of sensors and then adjust variables accordingly to correct for the disturbances or errors. The proportional term of a closed loop controller is manipulated using the proportional band setting, which makes a change to the output that is proportional to the error value.[3] The response can to manipulated through the use of equation (1) seen below:

$$\text{Proportional Grain (Kp)} = 100\% \,/\, \text{Proportional Band\%} \qquad (1)$$

When the proportional band is low and the proportional grain it high large changes in the output can be observed and the system may become unstable at very low band values. However, when the proportional band is high, and the grain is low the changes in the output become small and result in large input errors. This means that and very high band values the controller will be less sensitive and less responsive and the response time to disturbances may become very slow.

The integral term of a closed loop controller is proportional to the magnitude and the duration of the error and is often used in conjunction with the proportional term. The careful combination of these two terms makes the overall control more efficient when correcting for disturbance and errors. This integral term sums the instantaneous errors over time and gives accumulated offset that should have been corrected previously.[3] When the integral term is too long the correction of the offset will be very slow and when the integral term is too short this can cause for the system to become unstable. A third term, the derivative term, is also often add to closed loop controls as a way of slowing the rate of change of the controller output, however, is not used in this experiment.[3] This term is used to minimize the overshoot produced by the integral term and improves the controller's overall stability.

4. Results

In this experiment, the overall objective was to learn the fundamentals of process control by controlling the level of water in a process vessel and to determine the effect of P and I controllers on a system. A sump tank was filled up with water to the fill line and two different experiments were done; one was the manual control of the water level, experiment A, while the other was the proportional control of the water level, experiment C. In the first part of experiment A, the water was pumped into the vessel at a constant level of 38mm and then, the solenoid valve was opened. Once the solenoid valve opened, the level in the vessel started to decrease and once it had decreased to about 18mm, the pump speed was increased in order to increase the level of the vessel back to its original level. Once the original level was reached, the solenoid valve was closed. After the level increased by 20mm from the original level in the vessel, the pump speed was decreased to allow the vessel level to go back to 38mm. Figure 1, below, shows the change in the vessel levels caused by the increase and decrease of the pump speeds as well as the effects from the open and closed solenoid valve.

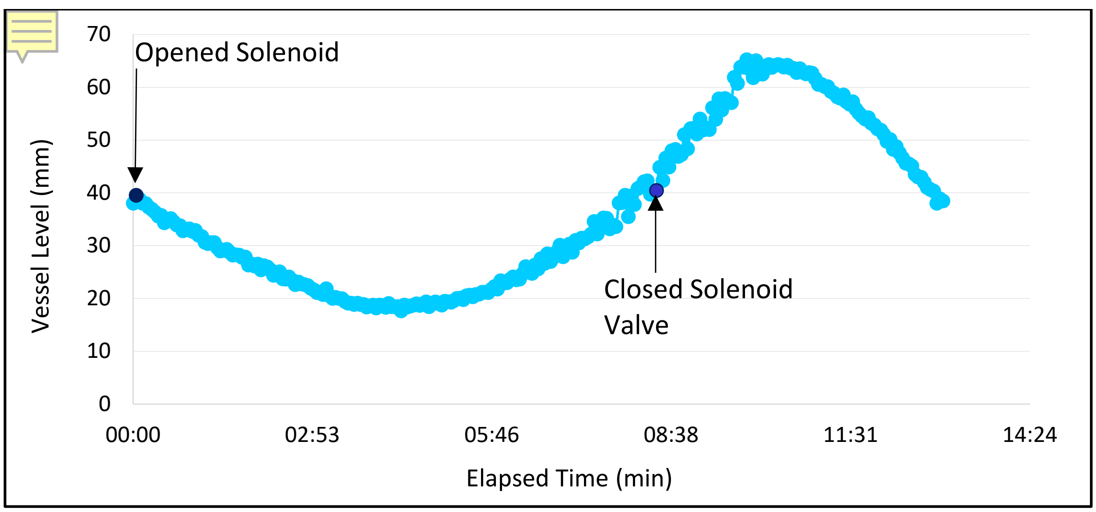

Figure 1: Experiment A, Effects of an Open/Closed Solenoid Valve

In the next part of experiment A, the pump speed was gradually increased to a speed until the level just started to rise. At that constant speed, the vessel was filled until the level was around 150mm. Then, the pump was stopped and the vessel was drained. Then, the pump speed was increased to 75% speed. Once the level reached 150mm, the pump was stopped and the vessel was drained. Figure 2, below, shows the different rates of filling at different pump speeds, but vessels draining at the same rate.

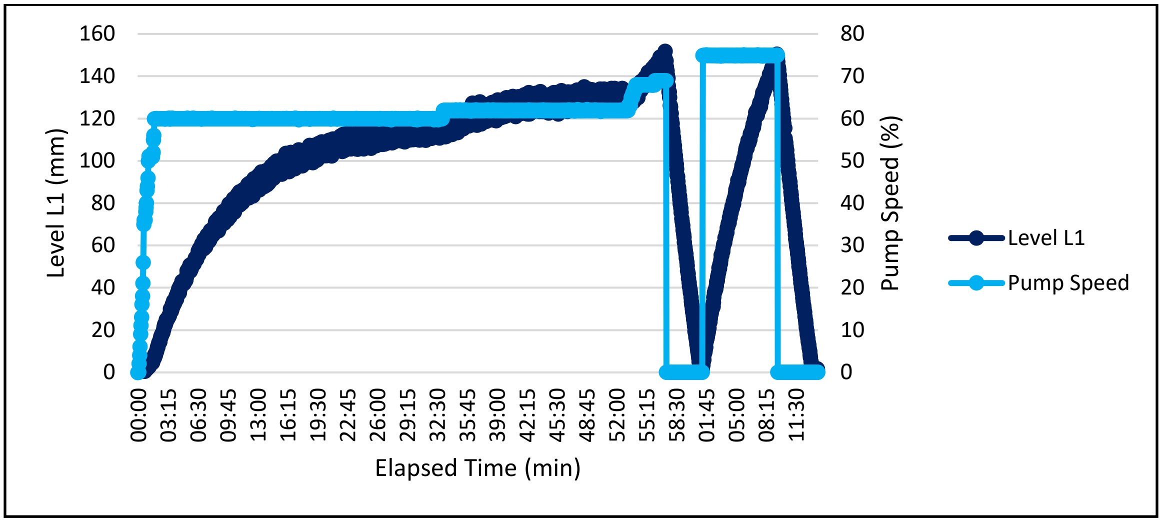

Figure 2: Experiment A, Effects of pump speed on the filling and draining of levels inside the vessel

In the first part of experiment C, P only controller, the mode of operation was in automatic mode. Different Proportional B and P settings were set and the process of stabilizing the level within the vessel was disturbed by opening and closing the solenoid valve. Due to the runs with the Proportional Band P of 200% and 100% not giving any values, the Proportional Band P of 50% was the first set of measurements that were taken. Figure 3, below, shows the changes in the water and the stability in response to the change in the P setting with a disturbance. Figures 6-10, in the appendix, shows the results of the Level L1 and the Controller output for each of the different P settings. From the figures 4-8, it shows a closer view of the levels that were in the vessel at the certain P setting and the variations in the levels may be due to the opening and closing of the solenoid valve.

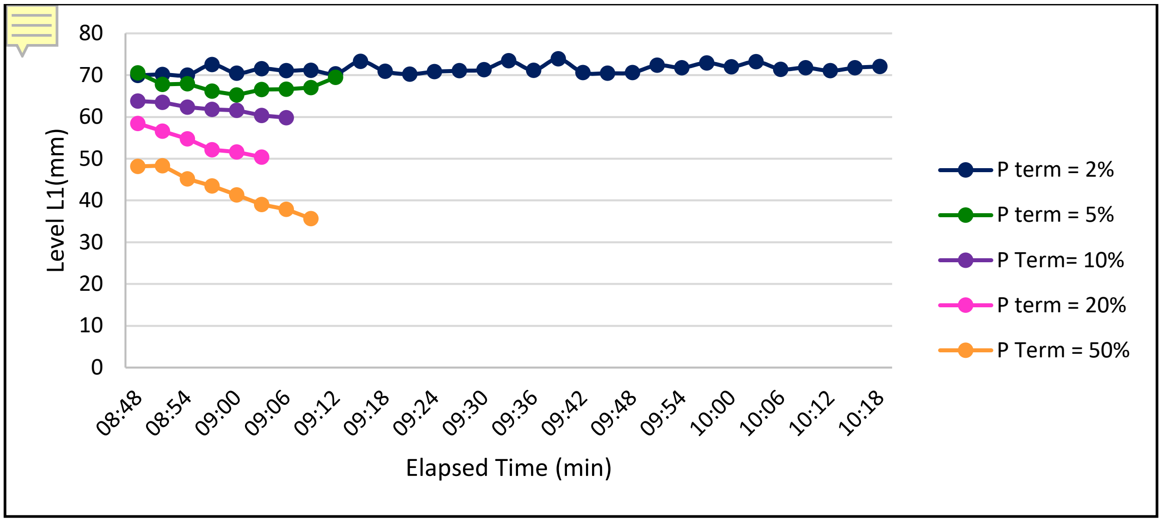

Figure 3: Experiment C, Change in water levels from the different P settings

In the second part of Experiment C, P + I Controller, the Proportional Band P was set to 20%, the I term was set to 100 seconds, and the set point was set to 75mm. The solenoid valve was opened to disturb the process and Figure 4 below shows the change in the water level and the pump speed from the change in the I setting as well as the disturbance from the solenoid valve. The pump speed was increased while trying to increase the water level to reach the 75mm and it decreased as it reached the level and as the solenoid valve was closed.

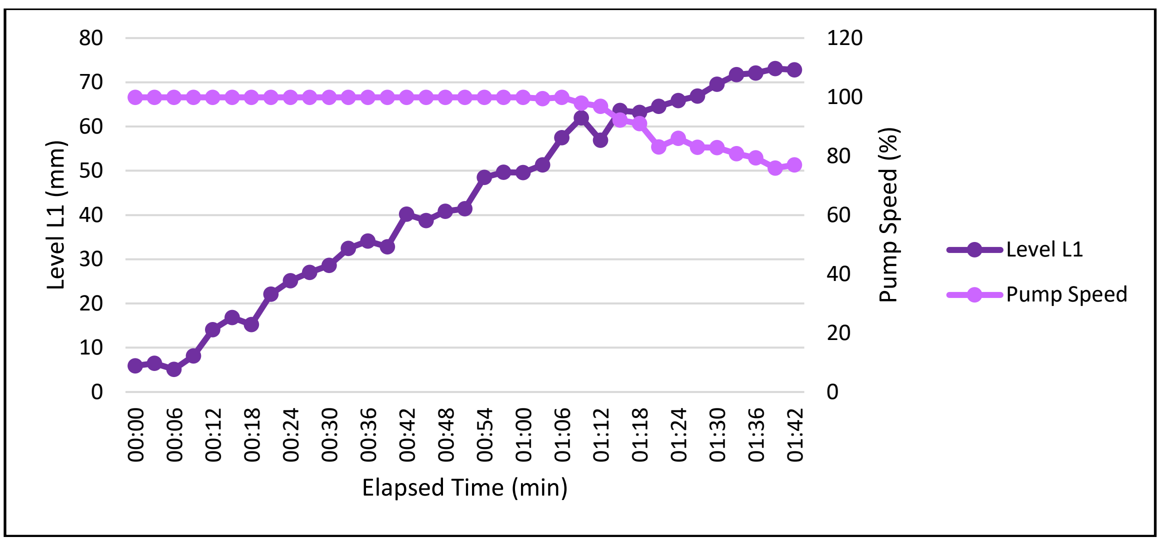

Figure 4: Vessel Level and Pump Speed as P is set to 20% and I term is set to 100

Then, until the level had become stable, the integral time setting was decreased and the disturbances were repeated. Figure 5, below, shows the difference of the water levels and the impacts of the disturbances as the integral term is decreased.

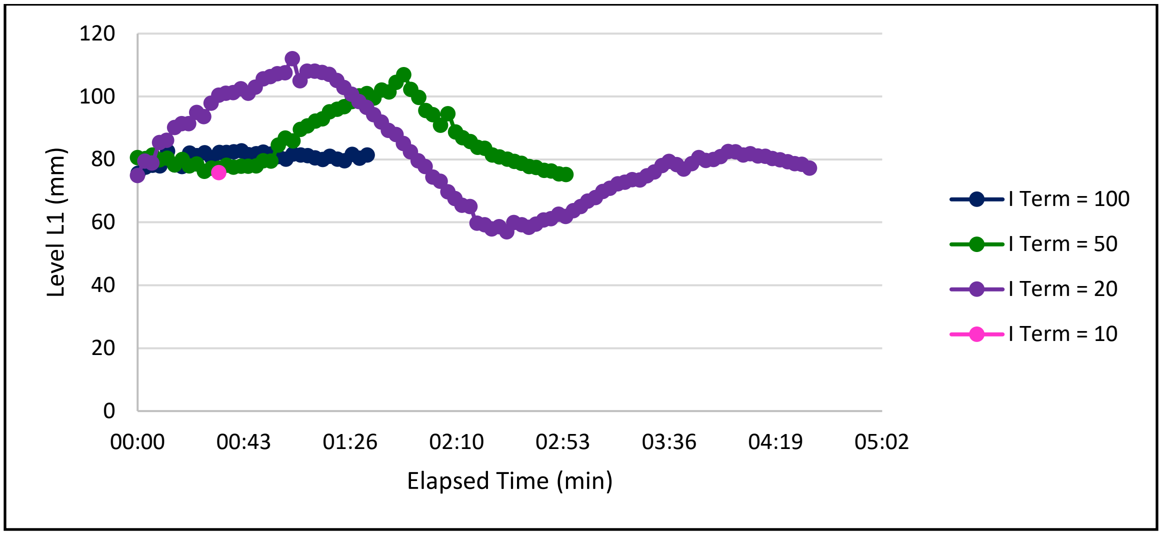

Figure 5:Level Response to Different Integral Time Settings

5. Discussion

In the first part of Experiment A, water is pumped in manually until a constant level of water is reached at 38mm as seen in Figure 1. After that the solenoid valve is opened and water is drained out of the sump tank to about 18mm. This result is anticipated because the pump isn't operating at a speed high enough to bring the flow rate going in to be equal than or more than the flow coming out of the sump tank. Another trend is seen in the next part of the graph where the pump speed is increased and with that the water level is also increased to the original height. This is also expected and agrees with literature because when the speed of the pump increases the liquid level increases as well, and vice versa. [5] Next, to observe the effects of the solenoid valve rather than the pump speed on the water level, the 8.38-minute mark and beyond on Figure 1 is analyzed. Here, the solenoid valve is closed off while keeping a constant pump speed. This will obviously allow for the water level to increase in the tank because the draining effect the solenoid valve had on the system is no longer present. At about 65mm, the pump speed is decreased and the level of water is seen to decrease, just like previously claimed in the theory section.

In the second half of Experiment A, the pump speed was gradually increased until the level of water just started rising and then kept at a constant speed. Then, as seen in Figure 2, the system is left alone to allow for the level of the water to rise to 150mm which takes about 55 minutes. The reason why it takes a long time for the tank to fill up is simply because the speed is set at such a low point that the level rise is bound to happen over a long period of time as observed. Once the specified level is reached, the tank is drained at a rather quick rate, especially compared to the time it took to fill up the tank. In this case, the pump is no longer filling the tank so the flow rate of the water going out is going to be the only factor contributing to the accumulation of the drained water; therefore allowing this process to happen much more quickly than filling up the tank. Next, the pump speed is increased to 75% of the original speed and the level is brought to the same height as before. Since this is done at a significantly higher speed it makes sense that the tank filled up much quicker than when at the original speed. Then once again the pump is turned off and the water is drained out from the same valve, with the same diameter, which results in the same exact trend as the first time the tank was drained.

In the first part of Experiment C, the P only controller was used to observe how the process control system will automatically react to disturbances in the trial. At first, this experiment was to be ran at the Proportional Band P of 200%. This percentage simply means that this process controller will make changes to the output proportional to the error value. [3] Since the system could not account for such a big error, the pump was not supplied with enough energy to pump water up. A run was also tried at a Proportional Band P of 100%, but also failed due to the same reasons mentioned previously. This observation is backed up by the theory stated previously that at very high band values the controller becomes less sensitive. For the sake of the experiment running, the percentage was brought down to 50%. In Figure 3, a band value of 50% proves to work very well in the system because despite any disturbances the controller took care of the issue immediately to the point where no disturbances can be seen on the graph. The same observation can be made for the 20% and 10% values. Once the band values were brought down to 5% and 2%, the disturbances created started to show up in the results as seen on Figure 3. The level of water wasn't being stabilized as it was previously at higher percentage values. This agrees with theory as well stating that at very low band values the controller becomes less sensitive.

In the second half of Experiment C, both the P + I controller was used. As previously stated, the P band was set 20%, the I term was set at 100 seconds and the set value at 75mm. Upon analyzing Figure 4, it is clear the P + I controller does a decent job at minimizing the effects of the disturbances because the water level increases quite linearly as seen on the graph. The system also worked well with these controllers at their set values because the system recognized that the desired level of water was approaching and decreased the pump speed in order to reach 75mm at about 100 seconds. This agrees with the theory stating that both these controllers working together would lead to the requested outcome in an efficient process. Once the desired level stabilized, the I term was manipulated to see its effects in the system. As seen in Figure 5, when the I value was decreased, the efficiency of the process was poorly affected. As the value was brought down, the longer it took for the system to reach the desired level compared to when the I term value was at 100. These observations agree with the theory that states the offset will be very slow when the integral term is too long and the fact the system will become unstable when the integral term is too short. It can be concluded that the P + I controllers must work together to create an effective process control system.

6. References

[1] Nolan, Dennis P. "Process Controls." *Handbook of Fire and Explosion Protection Engineering Principles (Second Edition)*, William Andrew Publishing, 13 Dec. 2010, www.sciencedirect.com/science/article/pii/B9781437778571000100.

[2] "Chapter 5." *Chemical Engineering Design*, by Raymond K. Sinnott and Gavin Towler, Second ed., Elsevier, 2020, pp. 251–261.

[3] Zdunek, Alan. *CHE 381:Unit Operations Laboratory-Process Control: Level Control*, Spring 2019, Department of Chemical Engineering, University of Illinois at Chicago, Chicago, IL.

[4] "Open-Loop System and Open-Loop Control Systems." *Basic Electronics Tutorials*, 4 Mar. 2018, www.electronics-tutorials.ws/systems/open-loop-system.html.

[5] Moran, Sean. "Pump Sizing: Bridging the Gap Between Theory and Practice." *AiChE.org*, Dec. 2016, www.aiche.org/sites/default/files/cep/20161238_r.pdf.

7. Appendix I: Data Tabulation/Graphs

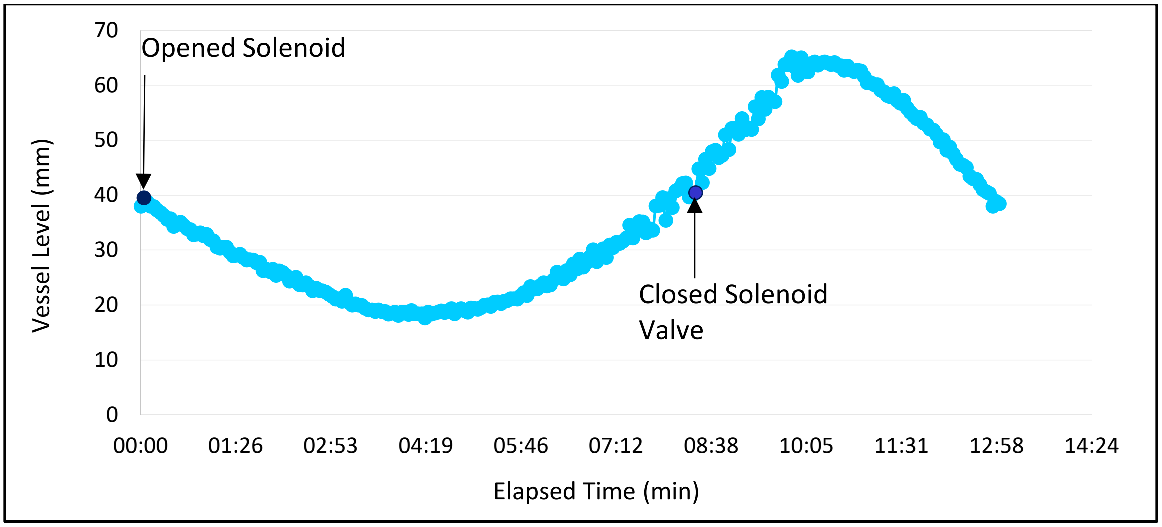

Figure 1: Experiment A, Effects of an Open/Closed Solenoid Valve

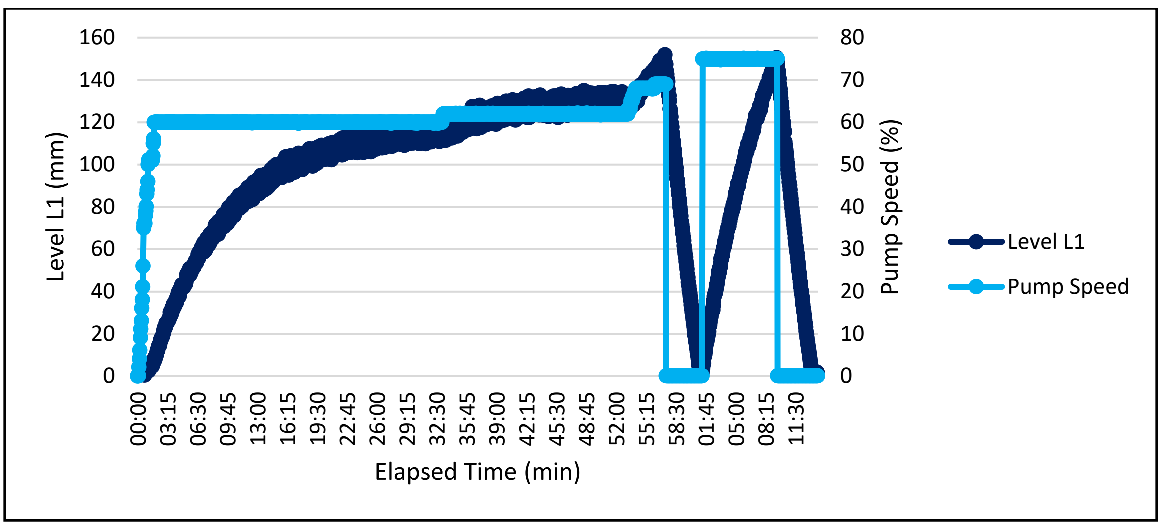

Figure 2: Experiment A, Effects of pump speed on the filling and draining of levels inside the vessel

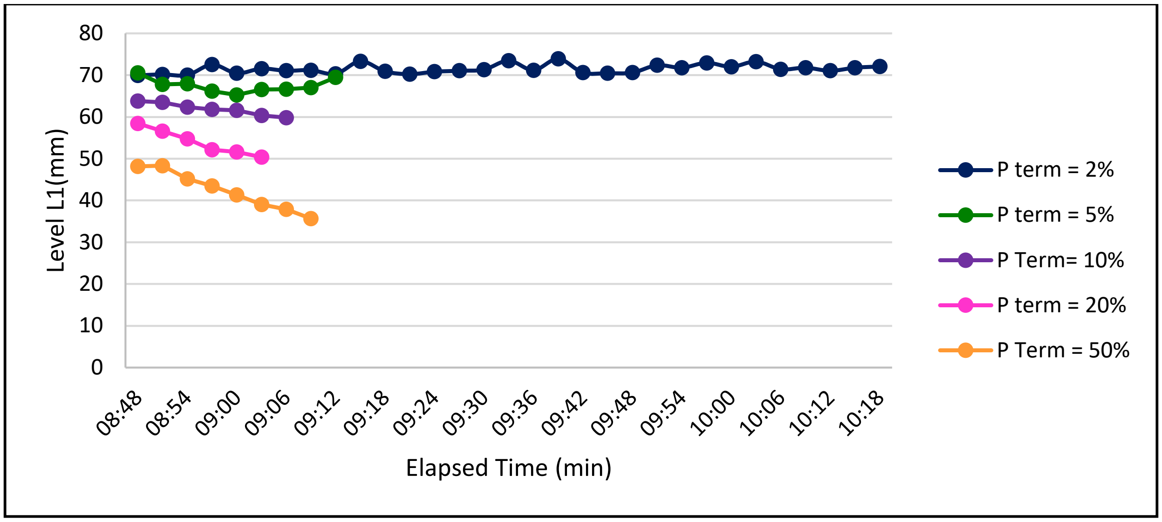

Figure 3: Experiment C, Change in water levels from the different P settings

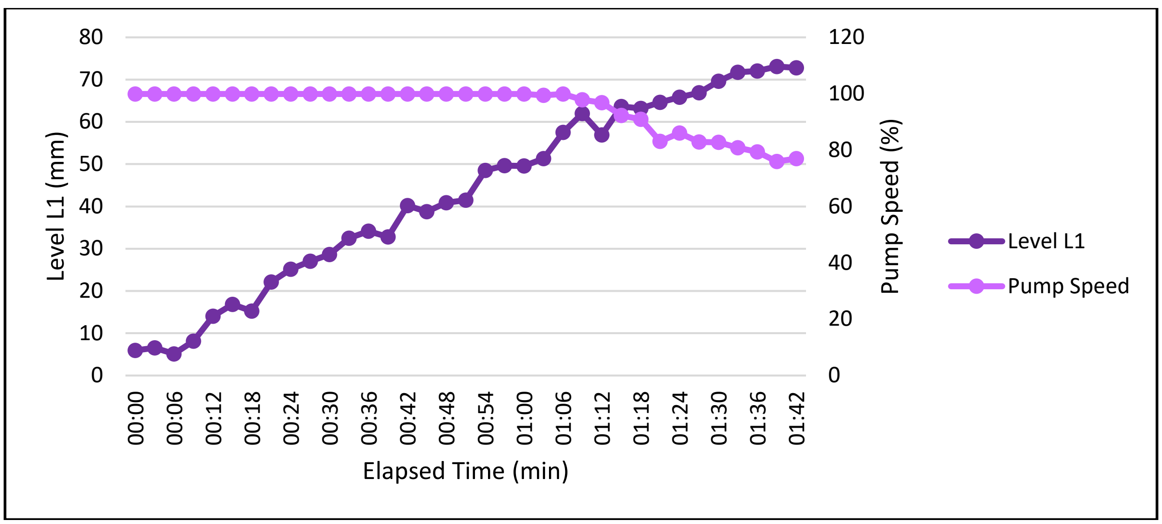

Figure 6: Vessel Level and Pump Speed as P is set to 20% and I term is set to 100

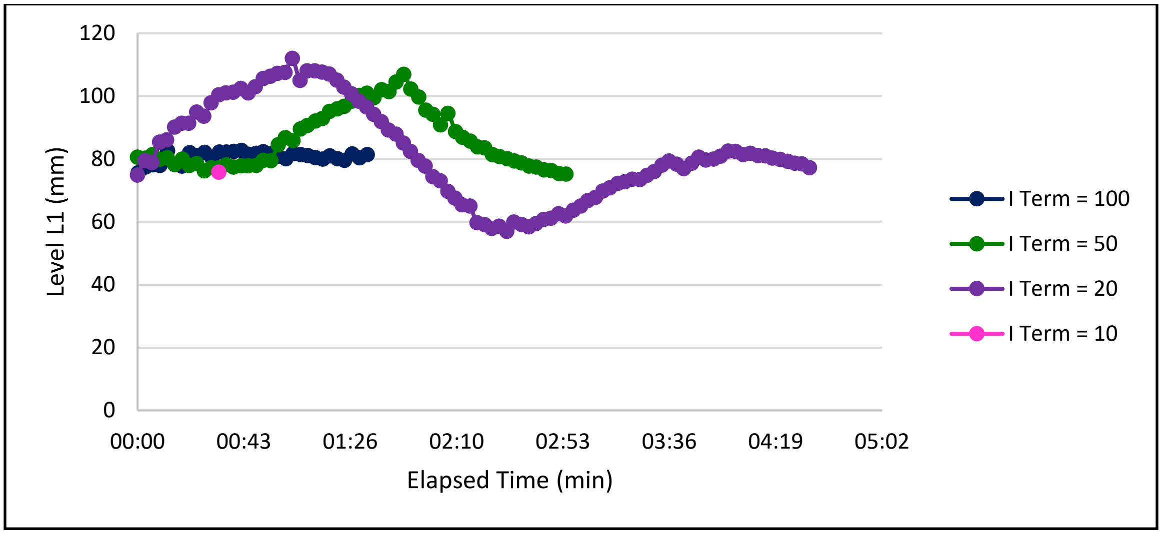

Figure 7:Level Response to Different Integral Time Settings

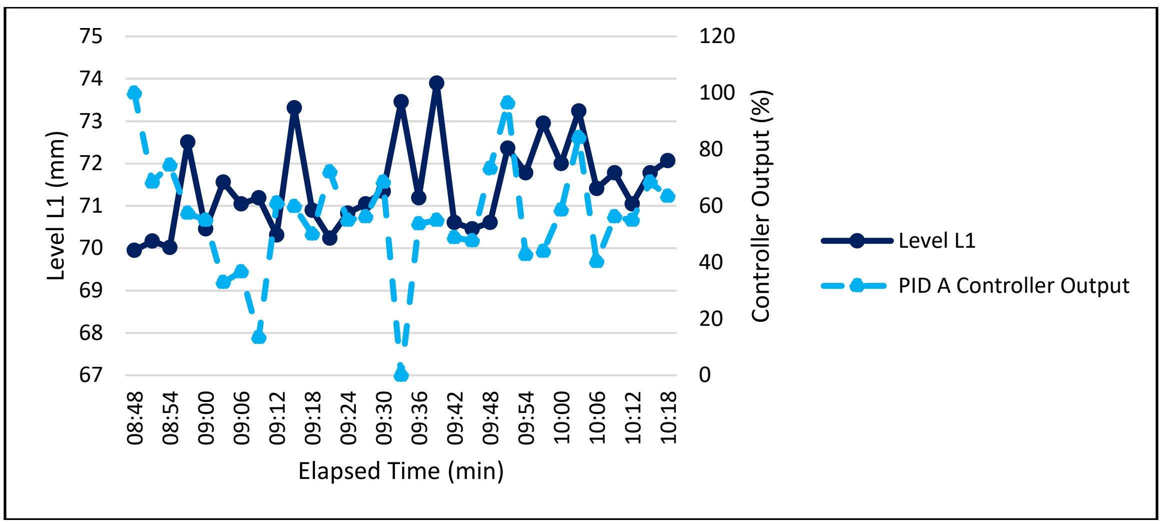

Figure 6: Comparison of Water Level and the Controller Output for Proportional Band P Set to 2%

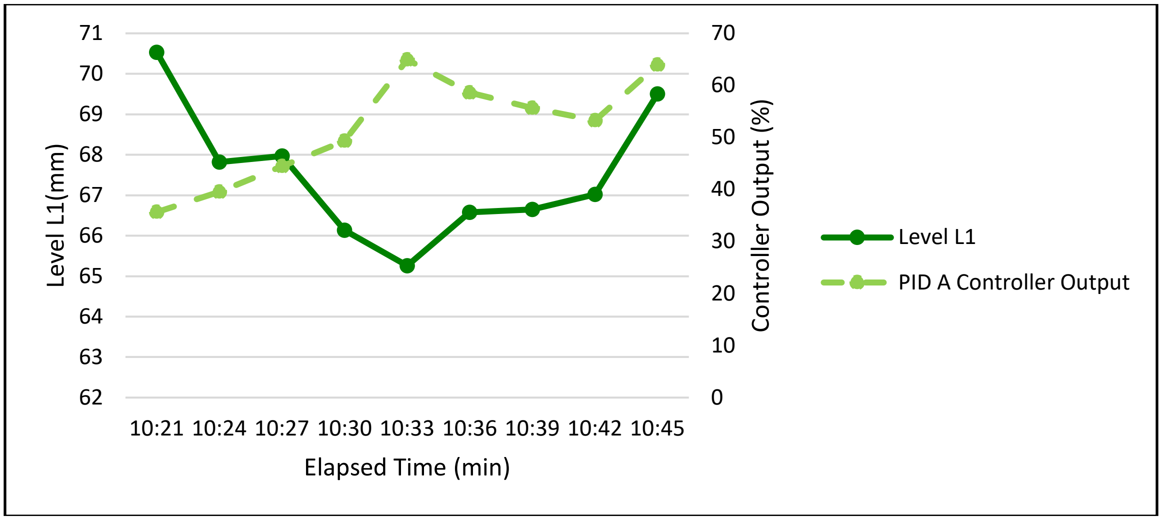

Figure 7: Comparison of Water Level and the Controller Output for Proportional Band P Set to 5%

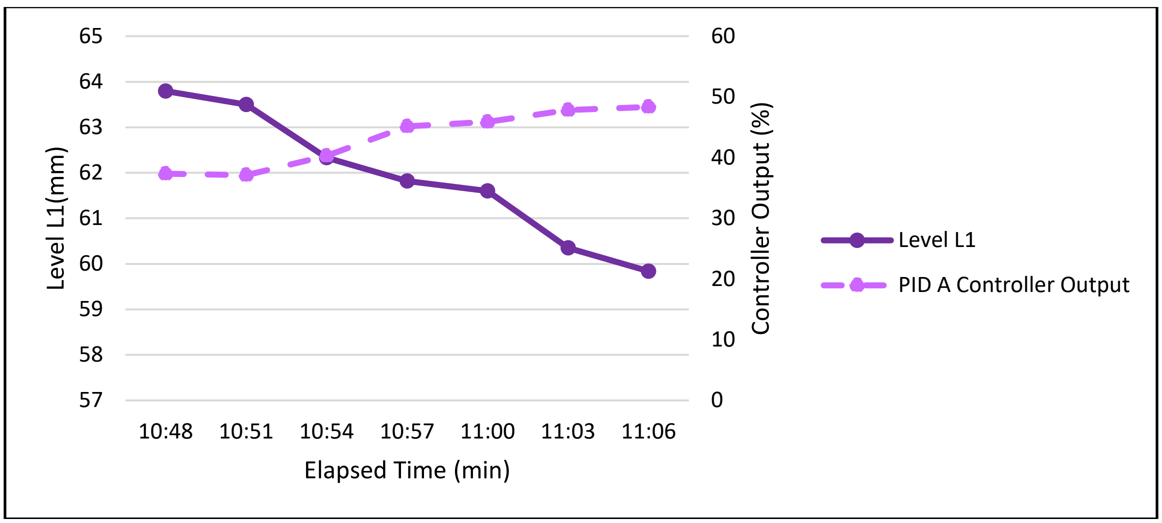

Figure 8: Comparison of Water Level and the Controller Output for Proportional Band P Set to 10%

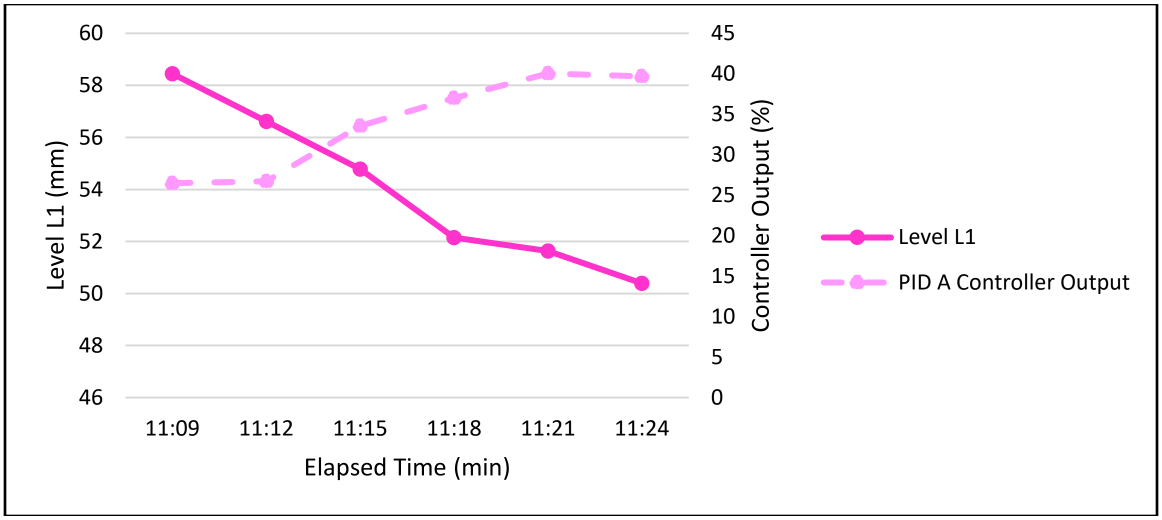

Figure 9: Comparison of Water Level and the Controller Output for Proportional Band P Set to 20%

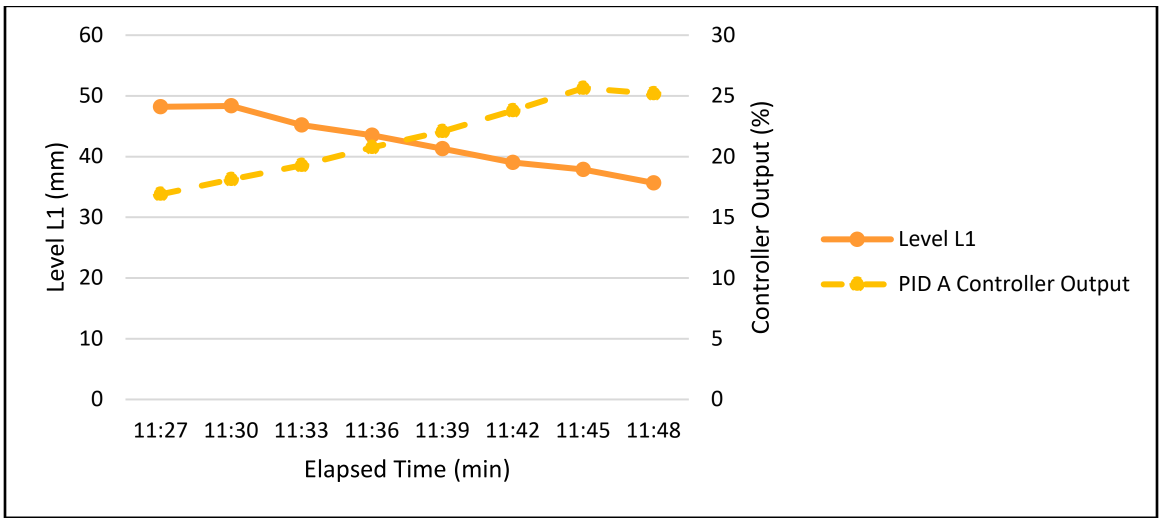

Figure 10: Comparison of Water Level and the Controller Output for Proportional Band P Set to 50%

8. Appendix II: Error Analysis

As the experiment is controlled via software, the only error will be human error on reading the results incorrectly. Another factor will be during low values of proportional band percentage where the process control oscillates rigorously. Finally, the apparatus is pre-recalibrated but is not compared to known values to ensure accuracy of the equipment.

9. Appendix III: Sample Calculations

In experiment C, the proportional gain is calculated by the following sample calculation:

$$Proportional\ Gain\ (Kp) = \frac{100\%}{Proportional\ Band\ \%}$$

$$Proportional\ Gain\ (Kp) = \frac{100\%}{50\%} = 2$$

10. Appendix IV: Individual Team Contributions

NAME: Diana Awimrin

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	Following Procedure and control levels going inside of the vessel
Pre-Lab Editing	10 mins	Typing and Editing
Experimental Objective (Prelab)	5 mins	The overall goal of the lab
Apparatus (prelab)	10mins	The set-up of the instruments and how they work
Materials & Supplies (prelab)	10 mins	List of the supplies needed
Procedure (prelab)	1	Detailed instructions of how to do the lab properly
Project Safety Evaluation (prelab)	30mins	Evaluating how the instrument could bring potential hazards
Final Lab Editing	1	Edited the final Report and proofread
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)	2	Discuss the results and relate it with the theory
References (final and/or prelab)	15 mins	Cited the sources that was used
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation	1	Added in information asked from the template PowerPoint
Total	12 hours 20 min	

NAME: Leena Daniel

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	Following Procedure and control levels going inside of the vessel
Pre-Lab Editing	10 mins	Typing and Editing
Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)	10 mins	List of the supplies used to make the solution
Procedure (prelab)	1	Detailed instructions of how to do the lab properly
Project Safety Evaluation (prelab)	5mins	Evaluating how the instrument could be potentially hazardous
Final Lab Editing	5 mins	Edited the final Report and proofread
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)	4 hours	Data Tables, Calculations for Plots and Results
Discussion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)	15 mins	Generated Tables used in Results
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation	20 mins	Added in information asked from the template PowerPoint
Total	12 hours and 20 mins	

NAME: Alice Guzik

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	Following Procedure and control levels going inside of the vessel
Pre-Lab Editing		
Experimental Objective (Prelab)		
Apparatus (prelab)	10mins	The set-up of the instruments and how it works
Materials & Supplies (prelab)	10 mins	List of the supplies used to make the solution
Procedure (prelab)	1	Detailed instructions of how to do the lab properly
Project Safety Evaluation (prelab)	30 mins	Evaluating how the instrument could be potentially hazardous
Final Lab Editing	1	Edited the final Report and proofread
Executive Summary (final lab)		
Introduction (Final lab)	1	Re-read lab manual and research online
Theory (Final Lab)	2	Researched Process Control
Results (final lab)		
Discussion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation	30 mins	Added in information asked from the template PowerPoint
Total	12 hours 20 mins	

NAME: Esteban Vazquez

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	Following Procedure and control levels going inside of the vessel
Pre-Lab Editing		
Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)	20 mins	List of the supplies used
Procedure (prelab)	1	Detailed instructions of how to do the lab properly
Project Safety Evaluation (prelab)	30mins	Evaluating how the instrument could be potentially hazardous
Final Lab Editing	1	Edited the final Report and proofread
Executive Summary (final lab)	1	Reviewed all the parts and consolidated it to a summary
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)		
References (final and/or prelab)	15 mins	Cited the sources that was used
Data Tabulation/Graphs (final)		
Error Analysis (final lab)	2	Analyzed the data given and calculated the regression and percent error.
Sample Calculations (final Lab)		
Power Point Presentation	30 mins	Added in information asked from the template PowerPoint
Total	12 hours 20 mins	